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A Framework to Promote Life Cycle Assessment (LCA) in Saudi Arabia's Oil and Gas Sector

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Abstract

To address the sustainability concerns of the oil and gas sector, it is important to reduce environmental impacts throughout its value chain. Understanding these environmental impacts can also help in developing the right policies needed to meet greenhouse gas emission targets. Life cycle assessment (LCA) is a comprehensive tool to quantify environmental impacts of products, processes, and systems across their entire life cycle. For more than the last thirty years, it has been used globally by various industries like construction, automotive, aerospace, etc. The objective of this paper is to present a theoretical framework to promote LCA in the Saudi Arabian oil and gas sector. This study relied on literature review to explore the current status of LCA in the Saudi Arabian oil and gas sector. Using the insights gained from literature, a framework consisting of three pillars—governance mechanisms, technical enablers, and competency building—is proposed. The framework encompasses the role of Saudi Arabian oil and gas industry stakeholders such as policymakers, relevant government agencies, and industry representatives in promoting the use of LCA. The adoption of this framework can accelerate the deployment of LCA studies, which in turn uncover environmental hotspots that are relevant to the local industry. This information can then be used to improve the sustainability of the entire oil and gas value chain in Saudi Arabia.

Introduction

In order to address the catastrophic consequences of global warming and climate change, active participation from all economic sectors is urgently needed. The oil and gas sector is the main driver of the economy in Saudi Arabia as it owns about 20% of world's prove oil reserves (OPEC, 2018). However, it is also a significant contributor to greenhouse gas emissions, water consumption, and other environmental issues that the country is facing (Taher & Hajjar, 2014). All activities related to the oil and gas industry like drilling, refining, and transportation lead to several environmental issues. Over the past decade, Saudi Arabia is also pursuing ambitious projects to diversify the economy and at the same time expand the oil and gas production and refining competency. As a result, the environmental footprint of the oil and gas sector is poised to grow significantly in coming years. Furthermore, current oil and gas infrastructure also contributes towards

environmental degradation through flaring, water usage, and land usage. So, there is urgent need to integrate robust sustainable practices across the entire value chain of oil and gas industry.

In addition to the current environmental impacts of the oil and gas industry, another important consideration for Saudi Arabia is the rapidly increasing population and resulting urban development. This growth will increase the demand for energy, transportation, housing, and industrial products that are linked to the oil and gas sector. Currently, the per capita carbon footprint of Saudi Arabia is one of the highest in the world. This situation creates a challenging energy-environment nexus that requires innovation and systematic measures to reduce the carbon footprint of the oil and gas industry. Historically, the oil and gas sector has prioritized economic efficiency and production growth over environmental concerns. Nevertheless, with growing environmental concern, the industry is moving rapidly to adopt sustainable practices. In this regard, Life Cycle Assessment (LCA) has emerged as a leading tool to quantify environmental impacts across the entire value chain of the oil and gas industry.

LCA is one of the most commonly used methodologies to quantify environmental impacts of products, processes, and systems. It examines environmental inputs and outputs, including energy consumption, emissions to air, water, and land, and waste generation (Baumann & Tillman, 2004; Hauschild et al., 1999; Pennington et al., 2004). For over the last thirty years, it has been extensively used in various industries to evaluate the sustainability of processes and systems (Unruh, 2015). The method is regulated by the ISO 14000 series of standards (International Organization for Standardization, 2006). LCA takes into account all stages of a product life cycle, including raw material extraction, transportation, processing, use, and recycling (Figure 1). It then helps to identify environmental hot spots — the stages that create the most damage to the environment in the life cycle. Identifying these hot spots is important to implement targeted actions to reduce the overall environmental impacts. LCA also helps to prevent burden shifting from one stage of the value chain to another. A process, for example, that reduces emissions during refining might increase toxicity during disposal. Such trade-offs are identified by LCA.

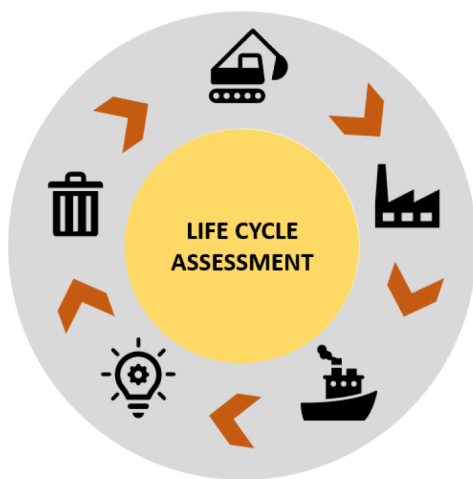


Figure 1—Life Cycle Assessment Stages

LCA offers many other strategic benefits. LCA can help in selecting environment-friendly technologies, optimizing energy and water use, and reducing emissions by optimizing process design. It can also play a vital role in circular economy strategies by identifying suitable waste management processes for disposing of sludges and retiring used equipment. Overall, LCA can enable oil and gas corporations to make informed and data-driven decisions to reduce carbon footprint and at the same time increase the efficiency of the processes. According to ISO standard (International Organization for Standardization, 2006) there are four steps in any given LCA study (Figure 2). Overall the process starts with developing the scope of a given study and the process ends with interpretation of results for improving sustainability. First of all, a clear goal

and scope of the study is established and system boundaries are defined. These system boundaries are then used to collect inventory related to material, energy, and emissions for a given process. After that various environmental impacts like global warming potential, ozone depletion potential, etc. are calculated using life cycle impact methods like CML. Finally, the results of the study are analyzed to propose recommendations for reducing the environmental burden of the process.

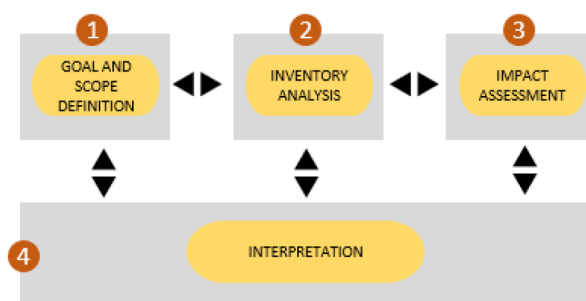


Figure 2—Steps for conducting Life Cycle Assessment (ISO, 2006)

This paper aims to develop a theoretical framework for integrating LCAs in Saudi Arabian oil and gas sector. First, the paper identifies the current status of LCA in the oil and gas sector by identifying relevant studies. Then it lists the key stakeholders, enablers, policies, and strategic practices needed to embed LCA in various oil and gas industry operations. Ultimately, this framework will provide guidelines to promote activities to achieve net zero commitments of Saudi oil and gas industry.

Methodology

This work relies on a literature review to first list the LCA studies related to the oil and gas sector in Saudi Arabia and then to develop a conceptual framework to promote LCA. Literature review is one of the most commonly used methods to explore the current status of a particular topic (Grant & Booth, 2009). These reviews help to delineate the state of the art in a given area, point out the gaps in research, and direct future research endeavors. In order to understand the current application of LCA in the Saudi Arabian oil and gas industry, a literature search was conducted using "Saudi Arabia," "LCA," and "Oil and Gas" as keywords. Scopus was the main database used for this search as it contains the most comprehensive collection of journals and conference proceedings related to sustainability, environmental science, and climate change. Additionally, a general search on the Internet was also performed using the same keywords. This additional search helps to find industry reports and policy papers that are usually not reported in scientific literatures.

As a result of the search, only four research papers and one industry report were found that mentioned the use of LCA to evaluate the environmental impacts related to the oil and gas industry in Saudi Arabia. A study compared the marine gas oil produced in Saudi Arabia with marine gas oil produced in the USA and LNG produced in Qatar (S. Hwang et al., 2019). The authors reported that LNG produced significantly less environmental impact as compared to marine oil. However, they further mentioned that loss of LNG during storage and transportation is a significant environmental issue that needs to be addressed. In a similar study, LCA was conducted to compare Saudi marine gas oil, Qatar's LNG, and hydrogen for meeting International Maritime Organization's 2050 target of greenhouse gas emissions (S. S. Hwang et al., 2020). The authors concluded that if hydrogen is produced with green energy then these targets can be met. In another study, using LCA it was found that loss of methane in the LNG supply chain can lower the environmental benefits of using LNG as compared to other fossil fuels (El-Houjeiri et al., 2019). The last paper compared the environmental impacts of various oil and gas pipeline repair techniques (Shaukat et al., 2022).

Given the far-reaching importance and huge carbon footprint of the oil and gas industry in Saudi Arabia, there is a major opportunity for the use of LCA and to expand the scope of its application in the complete

oil and gas value chain. Next, a literature review was conducted to find existing frameworks related to the application of life cycle assessment in various industries (El Haouat et al., 2025; Heijungs et al., 2010; Ijaz et al., 2024). By extracting key concepts, constructs, and themes, a relevant framework for the oil and gas industry was developed. Adopting this framework can help to accelerate the application of LCA in the Saudi oil and gas industry.

Framework

Here a framework comprising of three pillars: governance mechanisms, technology enablers, and competency building is presented. This framework is depicted in Figure 3 and each pillar is explained in detail in the following.

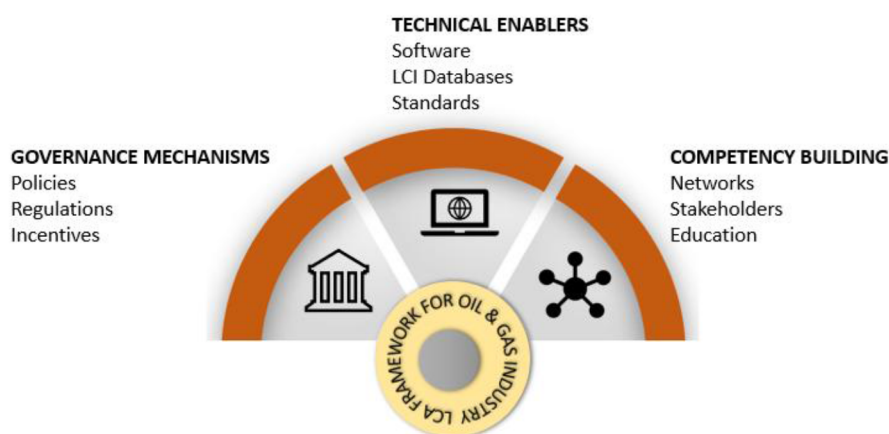


Figure 3—A Framework to Promote Life Cycle Assessment in Saudi Arabia

Governance Mechanisms

Government policies, regulations, and company standards for using LCA come under the governance aspect of the framework. In addition, sustainability reporting, financial incentives, and procurement policies are all related to the governance aspect of the framework. Government policy can encourage oil and gas companies to adopt LCA as a tool for assessing environmental impacts of both upstream and downstream projects. These policies can also encourage stakeholder collaboration and alignment to ensure sustainability becomes a paramount focus for a given project.

Saudi oil and gas operations' sustainability can be enhanced by policies that encourage implementing LCA. Environmental regulations, for instance, can advocate the use of LCA to assess environmental impacts of industrial projects and ensure minimal harmful impacts. Similarly, design standards can encourage the use of LCA during the design phase of oil and gas infrastructure and encourage the use of sustainable materials, technologies, and processes.

Another key driver of LCA will be mandatory corporate sustainability reporting. These reports can rely on LCA to provide transparent reporting of environmental impacts and encourage sustainability practices. Procurement policies for selecting materials, equipment, and services can integrate LCA to promote utilization of eco-friendly options. By implementing these measures, LCA can become an important tool to encourage sustainable practices. Nevertheless, for successful implementation, these policies must align with Saudi Arabian regulations and standards. Saudi Arabian Standards Organization (SASO) can play an important role in this regard by involving all stakeholders in delineating LCA policies.

Technical Enablers

A number of technical tools can help implement LCA. These tools include LCA software, relevant life cycle inventory data, methodology standardization, and process simulation. The availability of LCA software

facilitates the calculation of environmental impacts, conducting sensitivity and uncertainty analyses, and preparation of LCA documentation. However, the accuracy of LCA results heavily depends on the quality of data. Therefore, it is important to develop standard protocols to collect relevant data about material consumption, energy use, process efficiency, and emissions. Using data standards helps ensure consistency in calculating oil and gas project environmental impacts and allows for cross-comparisons between projects.

Apart from standard databases, developing local LCA data is a crucial step that can improve LCA results. Currently, most of the data used for calculating environmental impacts is based on global averages. Developing a local database using Saudi Arabian values can help improve the accuracy and relevance of LCA studies. Furthermore, LCA data can be integrated with digital twins and environmental assessment management systems. By implementing these enablers, LCA can be promoted within all ecosystems of the oil and gas industry. LCA tools, relevant data, and digital integration can drive the adaptation of LCA in the industry. However, while developing technical enablers is important, equally important is competency building.

Competency Building

Competency building aspect of the framework refers to the factors that facilitate the implementation of LCA in Saudi Arabian oil and gas industry. LCA is a rather involved sustainability assessment technique and requires effort for competency building. Competency building starts by involving key stakeholders like oil and gas engineers, oil and gas infrastructure designers, contractors, suppliers, and regulatory bodies. **Education** and **awareness** consisting of training programs, university-level courses, and conferences are also extremely necessary for local competency building. Of equal importance is collaboration and partnerships between these stakeholders to ensure shared understanding of underlying LCA principles. Moreover, the right incentives in terms of tax breaks and subsidies for entities that promote a sustainable oil and gas sector can also enhance competency building. However, competency building should align with the local norms and values to achieve their goals. For this reason, stakeholder development, education, and incentives should be tailored to Saudi Arabia's local industrial context.

However, it is to be noted that these aspects—governance mechanisms, technical enablers, and competency building - are interdependent and must be addressed together to achieve successful implementation of LCA, which will essentially lead to sustainable development in the oil and gas industry in Saudi Arabia. Once these enablers are adopted, this sector can become more sustainable; they will result in a reduction in environmental impact and the promotion of economic growth and social development.

Discussion

As discussed in the previous sections, the application of Life Cycle Assessment (LCA) in Saudi Arabia's oil and gas sector is still in its early stages, despite its significant potential for assessing the sustainability of oil and gas projects. Given the important role of the oil and gas industry in Saudi Arabia and its large carbon footprint, the adoption of LCA can support sustainability-driven decision-making. Several strategic steps can be taken to accelerate the adoption of LCA in this vital sector of the economy. These are discussed below.

Education and Training

Universities and institutions of higher education can play a pivotal role in promoting the use of LCA in Saudi Arabia's oil and gas sector. Historically, many LCA methodologies have been developed in academic settings. For example, the CML methodology was developed at Leiden University. In Saudi Arabia, universities can act as catalysts in enabling the adoption of LCA. LCA education can be integrated into both graduate and undergraduate engineering and environmental science programs. Similarly, professional development courses can be offered to promote awareness and practical knowledge of LCA. At present, a few universities have begun taking the initiative. Recently, KAUST has conducted conferences and workshops related to LCA, representing a positive step toward broader awareness and adoption.

LCA Networks

Currently, very few practitioners of LCA are in Saudi Arabia. Furthermore, LCA consulting firms also have a limited presence. To overcome these limitations, it is essential to initiate a collaborative network of experts that can undertake LCA studies for the oil and gas industry and provide training to professionals. Developing such a network will jumpstart the activities necessary for competency building. By doing so, Saudi Arabia will follow several advanced economies where such networks are already present and contributing to LCA competency building (Bjørn et al., 2013). One such example is a network started by the United Nations Environmental Program (UNEP) that connects LCA practitioners on a global level ([The Life Cycle Initiative | UNEP - UN Environment Program, n.d.](#)). These networks play an important role in developing standard practices and exchanging knowledge at national and international levels. In the beginning, a virtual network can be created that connects local experts and researchers to exchange expertise and ideas. Conducting a local conference related to LCA application in the oil and gas industry is another step that can create awareness of LCA. As the use of LCA grows in the oil and gas industry and other sectors of the economy, this virtual network can transform into a national-level entity that can help develop local standards, local life cycle inventories, and specialized LCA software.

Localized Life Cycle Inventory

Results of LCA studies depend upon the life cycle inventory databases used by LCA software. For this reason, it is very important to develop LCA databases that are representative of the local conditions of a given country. Currently, most of the LCA studies conducted for various sectors in Saudi Arabia rely on European databases. The results provided by these studies do not take into account local aspects of the Saudi industry. For example, most of the processes in Saudi Arabia rely on fossil fuel-generated electricity; however, the LCA database may be using a different mix of electricity. Such discrepancies in the data can lead to environmental impact calculations that are not very accurate. These studies provide quantitative results about environmental impacts. Several LCA practitioners have reported that a common challenge faced by them is the lack of reliable data (Lehmann et al., 2013). For these reasons, countries like Japan and India are developing their own local databases (Hunkeler et al., 1998; Narita et al., 2004). As the accuracy of LCA results within a given study is sensitive to local conditions, Saudi Arabia should commence the development of local databases. Doing so will not only enhance the accuracy of LCA results but also promote local LCA competency building.

Apart from inventory data, life cycle impact assessment (LCIA) methods also depend on the local environment and climate conditions. Factors like local population density, climate, and seasons affect the life cycle impact assessment methods. Therefore, it is also important to initiate efforts in this regard. SASO can take the lead in this effort. Developing local databases and impact assessment methods will greatly enhance the precision and accuracy of LCA results and thus improve the effectiveness of solutions developed by these studies.

Involving Adjacent Sectors

In order to fully realize the potential of LCA to help achieve sustainability targets in the oil and gas industry, it is important to promote the use of LCA in adjacent sectors as well. These include the oil and gas industry, petrochemicals and chemicals, transportation and logistics involving pipelines, rail services, etc., and the manufacturing of oil and gas exploration equipment. All of these can benefit sustainably from life cycle thinking. Similarly, hydrogen production and transportation are also a closely related sector that is seeing rapid development in Saudi Arabia. Conducting LCA studies on hydrogen production and transportation can help develop the most sustainable strategy for advancing this sector. A partnership between academia and various industrial sectors can assist in promoting the use of LCA across all sectors of the economy.

Conclusion

Provided that Saudi Arabia has one of the largest oil and gas industries in the world, there is an urgent need to accelerate the adoption of LCA as a sustainability assessment tool. This can help to precisely quantify the environmental footprint of various activities of the industry and at the same time point to the environmental hotspots that are responsible for the bulk of the impacts. Using this information, strategies can be devised to lower the overall emissions from the industry. Industry-wide use of LCA, however, will require developing expertise through collaboration, training, and partnerships. Similarly, developing local life cycle inventories can further enhance the quality of results of LCA studies and make them more reliable and beneficial. In conclusion, this paper points to very limited application of LCA in Saudi Arabia's oil and gas industry and at the same time provides a framework to fully realize the potential of LCA. As the country targets net zero emission by 2060, LCA can play an important role in delineating the roadmap for a greener oil industry in future. Promotion of LCA training in educational institutions, developing local databases, and establishing local and global networks of practitioners will pave the way for a more sustainable Saudi oil and gas sector.

References

- Baumann, H., & Tillman, A.-M. (2004). The Hitch Hiker's Guide to LCA. In *Studentlitteratur Lund*.
- Bjørn, A., Owsianiak, M., Laurent, A., Molin, C., Westh, T. B., & Hauschild, M. Z. (2013). Mapping and characterization of LCA networks. *International Journal of Life Cycle Assessment*, **18**(4). <https://doi.org/10.1007/s11367-012-0524-6>
- El Haouat, Z., Essalih, S., Bennouna, F., & Amegouz, D. (2025). Development of a Global Framework for an Integrated Life Cycle Assessment (LCA) Model in Quality, Safety and Environmental (QSE) Management Systems: Improving Environmental, Social and Economic Sustainability Performance. *Sustainability (Switzerland)*, **17**(8). <https://doi.org/10.3390/su17083521>
- El-Houjeiri, H., Monfort, J. C., Bouchard, J., & Przesmitzki, S. (2019). Life Cycle Assessment of Greenhouse Gas Emissions from Marine Fuels: A Case Study of Saudi Crude Oil versus Natural Gas in Different Global Regions. *Journal of Industrial Ecology*, **23**(2). <https://doi.org/10.1111/jiec.12751>
- Grant, M. J., & Booth, A. (2009). A typology of reviews: An analysis of 14 review types and associated methodologies. In *Health Information and Libraries Journal* (Vol. **26**, Issue 2). <https://doi.org/10.1111/j.1471-1842.2009.00848.x>
- Hauschild, M., Wenzel, H., & Alting, L. (1999). Life cycle design - a route to the sustainable industrial culture. *CIRP Annals - Manufacturing Technology*, **48**(1). [https://doi.org/10.1016/S0007-8506\(07\)63210-7](https://doi.org/10.1016/S0007-8506(07)63210-7)
- Heijungs, R., Huppes, G., & Guinée, J. B. (2010). Life cycle assessment and sustainability analysis of products, materials and technologies. Toward a scientific framework for sustainability life cycle analysis. *Polymer Degradation and Stability*, **95**(3), 422–428. <https://doi.org/10.1016/j.polymdegradstab.2009.11.010>
- Hunkeler, D., Yasui, I., & Yamamoto, R. (1998). LCA in Japan: Policy and progress. I. *International Journal of Life Cycle Assessment* (Vol. **3**, Issue 3). <https://doi.org/10.1007/BF02978820>
- Hwang, S., Jeong, B., Jung, K., Kim, M., & Zhou, P. (2019). Life cycle assessment of lng fueled vessel in domestic services. *Journal of Marine Science and Engineering*, **7**(10). <https://doi.org/10.3390/jmse7100359>
- Hwang, S. S., Gil, S. J., Lee, G. N., Won Lee, J., Park, H., Jung, K. H., & Suh, S. B. (2020). Life cycle assessment of alternative ship fuels for coastal ferry operating in republic of Korea. *Journal of Marine Science and Engineering*, **8**(9). <https://doi.org/10.3390/JMSE8090660>
- Ijaz, T., Qazalbash, A. H. R., Razzaq, A. A., Rafique, M. Z., Khan, M. A., & Jamshaid, S. H. (2024). Development of green manufacturing implementation framework based on life-cycle assessment. *Sustainable Environment Research*, **34**(1). <https://doi.org/10.1186/s42834-024-00229-7>
- International Organization for Standardization. (2006). *ISO 14040 Environmental management--life cycle assessment--requirements and guidelines*.
- Lehmann, A., Zschieschang, E., Traverso, M., Finkbeiner, M., & Schebek, L. (2013). Social aspects for sustainability assessment of technologies - Challenges for social life cycle assessment (SLCA). *International Journal of Life Cycle Assessment*, **18**(8). <https://doi.org/10.1007/s11367-013-0594-0>
- Narita, N., Nakahara, Y., Morimoto, M., Aoki, R., & Suda, S. (2004). Current LCA database development in Japan -results of the LCA project. *The International Journal of Life Cycle Assessment*, **9**(6). <https://doi.org/10.1007/bf02979077>
- OPEC. (2018). OPEC Annual Statistical Bulletin 2018. *Organization of the Petroleum Exporting Countries*.
- Pennington, D. W., Potting, J., Finnveden, G., Lindeijer, E., Jolliet, O., Rydberg, T., & Rebitzer, G. (2004). Life cycle assessment Part 2: Current impact assessment practice. In *Environment International* (Vol. **30**, Issue 5). <https://doi.org/10.1016/j.envint.2003.12.009>

- Shaukat, M. M., Ashraf, F., Asif, M., Pashah, S., & Makawi, M. (2022). Environmental Impact Analysis of Oil and Gas Pipe Repair Techniques Using Life Cycle Assessment (LCA). *Sustainability (Switzerland)*, **14**(15). <https://doi.org/10.3390/su14159499>
- Taher, N., & Hajjar, B. (2014). Environmental Concerns and Policies in Saudi Arabia. In *Energy and Environment in Saudi Arabia: Concerns & Opportunities* (pp. 27–51). Springer International Publishing. https://doi.org/10.1007/978-3-319-02982-5_2
- The Life Cycle Initiative | UNEP - UN Environment Programme. (n.d.). Retrieved June 12, 2022, from <https://www.unep.org/explore-topics/resource-efficiency/what-we-do/life-cycle-initiative>
- Unruh, G. (2015). Strategic Sustainability Uses of Life- Cycle Analysis. *MITSloan Management Review*, 22–25.